

Intelligent Decision Support System for Sri Lankan Tea Estate Management

R26-IT-118

Project Proposal Report

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Sri Lanka Institute of Information Technology

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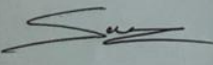
Declaration

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Abstract

Tea grading is a critical activity in Sri Lankan tea factories because the identified grade directly influences quality assurance, pricing, production planning, and market confidence. In many factories, tea sorting and grading still depend on manual visual assessment supported by operator experience and sieve-based separation. Although this practice is widely used, the final grading outcome can vary due to human subjectivity, fatigue, subtle differences among batches, and the absence of structured digital decision support. These limitations reduce consistency and make it difficult to generate reliable batch-level analytics for management use.

This research proposes an AI-Based Tea Sorting and Grading Automation module as a component of an intelligent decision support system for Sri Lankan tea estate management. The proposed solution uses image-based analysis and Convolutional Neural Networks (CNNs) to classify tea grades such as Dust, Pekoe, BOP, and FBOP using visible characteristics including size, texture, color, and broken-leaf ratio. A tea-grade image dataset will be developed using samples collected under controlled conditions from a factory-oriented environment. The trained model will then be evaluated using standard machine learning metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis.

Beyond single-image classification, the proposed system will support batch-wise grade summaries, confidence-aware outputs, and daily production reports that can help quality control officers and factory managers make more consistent decisions. The study will also compare AI-based grading outcomes with manual sorting and grading results in order to assess practical usefulness, reliability, and alignment with existing workflows.

By combining CNN-based image classification, reporting, and decision support, this research contributes a tea-specific grading framework that is suitable for Sri Lankan industrial conditions. The expected outcome is a context-aware grading support system that improves grading consistency, reduces subjectivity, and strengthens digital quality monitoring in tea manufacturing environments.

From a social and commercial perspective, the proposed system has the potential to support fair and consistent quality assessment across tea factories, reduce operator fatigue-related grading errors, and provide small and medium-sized factories with affordable AI-assisted quality tools. Commercially, the solution can be extended into a subscription-based service, contributing to the broader digital transformation of Sri Lanka's tea export industry and strengthening the country's competitive position in global tea markets.

Keywords

Tea grading, tea sorting, convolutional neural network, computer vision, image classification, decision support system, quality control, Sri Lankan tea industry

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List of Abbreviations

AI Artificial Intelligence

BOP Broken Orange Pekoe

CNN Convolutional Neural Network

DSS Decision Support System

FBOP Flowery Broken Orange Pekoe

ML Machine Learning

SDG Sustainable Development Goal

SLIIT Sri Lanka Institute of Information Technology

SLTB Sri Lanka Tea Board

UI User Interface

VPS Virtual Private Server

1. INTRODUCTION

1.1 Background and Context

The tea industry remains one of Sri Lanka's most important agricultural and export sectors. The quality and uniformity of made tea influence auction prices, buyer trust, brand reputation, and long-term competitiveness. Within the tea manufacturing process, sorting and grading are therefore not minor support activities but key quality-control steps that determine how the product is described, stored, marketed, and sold.

In practice, tea sorting and grading are commonly supported by a combination of mechanical separation, sieve-based processes, and human visual judgment. Trained factory personnel inspect tea particles and identify expected grades based on appearance, particle size, texture, color, and the amount of broken-leaf content. However, even experienced operators may produce slightly different judgments when the visual distinction among grades is subtle or when batch quality is mixed. These differences create inconsistency in reporting and may affect downstream production decisions.

Manual grading also creates operational challenges. The process can be time-consuming, subjective, and difficult to standardize across shifts, teams, and factories. When there is no digital mechanism for storing classification evidence, factories also lose the opportunity to analyze historical grading patterns, uncertain cases, and recurring quality variations. As a result, management decisions are often based on isolated observations rather than structured data.

Recent advances in artificial intelligence, especially deep learning and computer vision, provide a practical opportunity to improve this situation. Convolutional Neural Networks (CNNs) have shown strong performance in image-based recognition tasks because they can automatically learn meaningful visual features from raw image data. This makes CNNs suitable for tea grading, where visible particle characteristics are central to classification.

This research proposes an AI-Based Tea Sorting and Grading Automation system that classifies tea grades using tea sample images collected in controlled factory-related settings. The system is designed not as a replacement for all human work, but as an intelligent decision support tool that can improve consistency, support batch-level review, and provide production reports for quality monitoring.

Statistical Context: Sri Lanka is the world's fourth-largest tea producer, contributing approximately 7–8% of global tea exports and generating over USD 1.25 billion in annual export revenue (Sri Lanka Tea Board, 2023). The country operates more than 600 registered tea factories producing over 300 million kilograms of made tea annually. Studies indicate that grading inconsistency between shifts and operators can account for up to 10–15% variance in auction valuation for borderline-grade batches, representing a significant commercial risk across the supply chain.

SDG Alignment: This research aligns with United Nations Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by introducing an AI-assisted digital tool that modernizes tea manufacturing quality control. It also contributes to SDG 8 (Decent Work and Economic Growth) by improving productivity and supporting the economic sustainability of Sri Lanka's tea sector.

Furthermore, the adoption of data-driven quality monitoring supports SDG 12 (Responsible Consumption and Production) through more consistent and waste-reducing production practices.

Research Cluster Alignment: This proposal falls within the Intelligent Systems and Data Science research cluster, specifically addressing applied machine learning, computer vision, and decision support systems in an agro-industrial context. It is also relevant to the Digital Transformation and Smart Agriculture cluster due to its focus on deploying AI solutions within a traditional Sri Lankan industrial domain.

1.2 Literature Review

Artificial intelligence and machine learning have become widely adopted in agriculture and food-processing environments to support classification, defect detection, and quality monitoring. Kamilaris and Prenafeta-Boldú [4] conducted a comprehensive survey of deep learning applications in agriculture, demonstrating that CNN-based methods consistently outperform traditional handcrafted feature approaches across diverse crop-quality and classification tasks. In image-based grading tasks specifically, Blasco et al. [3] demonstrated that machine vision systems can automate fruit quality grading with high accuracy by leveraging visual features such as size, colour, and surface texture — characteristics directly comparable to those used in tea grading. These foundational works establish that visual AI models are technically mature enough to be applied in industrial food-quality settings.

Within the tea domain, Mohanty et al. [1] showed that deep learning models can identify subtle visual patterns in plant imagery that are beyond the reliable detection threshold of human inspectors, providing an early benchmark for image-based agricultural classification. Extending this to manufactured tea, Zhang et al. [2] applied deep convolutional neural networks combined with transfer learning to automate tea quality identification, reporting strong classification performance on tea appearance attributes. Similarly, Wu et al. [6] reviewed the state of deep learning for tea quality monitoring and concluded that CNN-based models are well suited for tea-specific recognition tasks when domain-relevant training data is provided. The Tea Research Institute of Sri Lanka [7] and the Sri Lanka Tea Board [8] further document the visual grade categories used in Sri Lankan production — including Dust, Pekoe, BOP, and FBOP — confirming that grade distinctions rely on measurable image-level features such as particle size, colour tone, and broken-leaf ratio.

Research has also addressed specific process stages within tea manufacturing. Kimutai et al. [9] developed a CNN-based optimum fermentation detection model for tea, establishing that deep learning can be meaningfully applied at individual factory process stages rather than only in field or agricultural settings. More recently, Yi et al. [10] proposed a multi-scale tea bud grading detection framework capable of operating in complex environments, demonstrating that grading-related visual diversity can be addressed through multi-resolution CNN architectures. On the sorting side, Zhang et al. [5] designed and evaluated a deep-learning-based fresh tea leaf sorting system, reporting that automated sorting using image classification models can reduce manual effort and improve throughput consistency in tea processing lines.

However, tea grading in a factory environment presents a fundamentally different challenge from disease recognition or bud detection in plantations. Manufactured tea grades can share overlapping visual features, and factory-grade classification must account for particle size distribution, colour tone, texture, and broken-leaf ratio simultaneously [7, 8]. A model intended for production-floor use must therefore be trained on tea-grade images that reflect real operational conditions rather than field or general agricultural image datasets [4, 6]. No existing work directly addresses the construction and evaluation of a Sri Lankan factory-grade image dataset across Dust, Pekoe, BOP, and FBOP categories under controlled capture conditions.

A further limitation across reviewed studies is that most published models report classification accuracy metrics only [1, 2, 9, 10]. In real production settings, quality-control staff additionally require confidence scores, batch-wise summaries, mismatch flags, and daily production reports to make grading decisions and maintain audit records [3, 5]. Systems that omit these decision-support components may achieve strong model metrics in isolation but offer limited practical value when deployed in factory workflows. The literature therefore supports the use of CNNs for visual grading but also highlights the need for a tea-specific, factory-oriented decision-support framework that integrates classification with operational reporting — a gap that this proposal directly addresses.

1.3 Research Gap

Although computer vision and CNN-based classification are widely researched, there is limited work specifically focused on Sri Lankan tea grading in factory environments. Most related studies either address a different agricultural problem or use datasets and evaluation settings that do not reflect the local grading context of made tea.

First, there is a clear gap in locally relevant image datasets for tea grades such as Dust, Pekoe, BOP, and FBOP captured under controlled but realistic factory-related conditions. Without such data, model development remains disconnected from the actual visual variation encountered during production.

Second, existing research rarely combines classification with decision support outputs such as batch summaries, confidence-aware reports, and comparison with manual grading. These practical features are essential for factory adoption because they help users trust and interpret model predictions.

Third, most tea-related AI studies focus on detection or prediction problems outside the sorting room, such as disease recognition, yield estimation, or field analytics. The grading stage remains underexplored despite its strong impact on quality assurance and commercial value.

This research addresses these gaps by proposing a tea-specific grading support system for Sri Lankan conditions, using CNN-based image analysis, structured reporting, and manual-versus-AI comparison within one coherent proposal.

2. OBJECTIVE

2.1 Main Objective

To design and evaluate an AI-Based Tea Sorting and Grading Automation system that classifies Sri Lankan tea grades from image data and supports factory-level decision-making through batch-wise grading analysis and reporting.

2.2 Specific Objectives

- To collect and label a tea-grade image dataset representing categories such as Dust, Pekoe, BOP, and FBOP from a factory-related environment.
- To preprocess tea images under controlled conditions in order to reduce noise and improve model consistency.
- To train and evaluate a CNN-based model for tea-grade classification using visible characteristics such as size, texture, color, and broken-leaf ratio.
- To generate batch-wise grade classification outputs and daily production reports based on model predictions.
- To compare the performance of AI-assisted grading with manual sorting practices in terms of accuracy, consistency, and usability.
- To assess the role of AI-assisted grading as a decision support mechanism rather than a fully autonomous industrial replacement.

3. METHODOLOGY

3.1 Research Approach

This study follows a design-and-development research approach with an experimental evaluation component. The research begins by identifying a practical quality-control problem in tea grading, then moves toward the design of an intelligent support solution, dataset preparation, model development, system integration, and performance validation.

The work is both technical and applied. Technically, it involves dataset creation, preprocessing, CNN model training, and evaluation. Practically, it examines whether the outputs of the system can assist quality-control staff in grading activities and report generation. The proposed solution is therefore positioned as a prototype decision-support system that can later be extended into a wider factory deployment.

3.2 Technical Methods and Algorithms

The core analytical method used in the research is a Convolutional Neural Network (CNN) for image classification. CNNs are appropriate for this problem because they automatically learn visual patterns from labeled training images and can distinguish among visually similar classes when provided with enough representative data.

The technical pipeline will include image acquisition, image labeling, preprocessing, data splitting, model training, hyperparameter tuning, evaluation, and result interpretation. Preprocessing is expected to include image resizing, normalization, controlled background handling, and data augmentation. The study may also compare a baseline custom CNN with transfer-learning models such as MobileNet, ResNet, or EfficientNet when beneficial.

For report generation, predicted labels and confidence scores will be aggregated at batch level. The system will summarize the number of samples classified per grade, highlight uncertain cases, and produce simple daily production outputs suitable for supervisor and management review.

3.3 Tools and Platforms

The proposed prototype can be implemented using Python as the primary development language. TensorFlow or PyTorch can be used to develop the CNN models, while OpenCV can support preprocessing operations. Experiments may be performed in Jupyter Notebook or Google Colab depending on available computing resources.

A lightweight web interface can be developed using Flask, FastAPI, or Django, with HTML, CSS, and JavaScript used for the front-end. Classification results, batch summaries, and audit records can be stored using MySQL or PostgreSQL. Visualization of analytics and reports may be produced using Matplotlib, Plotly, or similar tools.

3.4 Data Collection and Ethical Considerations

The dataset will be built using tea-grade samples collected from factory-oriented environments under controlled lighting and background conditions. Images will be labeled based on verified grade

categories with guidance from knowledgeable personnel where available. Metadata such as date, batch identifier, source location, and operator remarks can also be recorded when possible.

The study does not involve sensitive personal data, but factory permissions and operational confidentiality remain important. Permission should be obtained before collecting any image data from real production settings. If manual grading results are used for comparison, these records should be handled carefully and only for the approved research purpose. The study should also clarify that the proposed system provides decision support and does not make unrestricted industrial decisions without human oversight.

3.5 Validation Strategy and Performance Metrics

Model performance will be evaluated using standard classification metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis. These metrics will help determine how well the model differentiates among tea grades and where misclassification is likely to occur.

Practical validation will also be carried out by comparing AI-generated classifications with manual grading decisions. This comparison will focus on agreement rate, consistency across repeated batches, usefulness of confidence scores, and value of the generated daily reports. The combination of technical and operational evaluation strengthens the relevance of the final system.

Figure 3.1 shows the CNN tea grade model evaluation summary, presenting the key performance metrics obtained during model testing across the four grade categories.

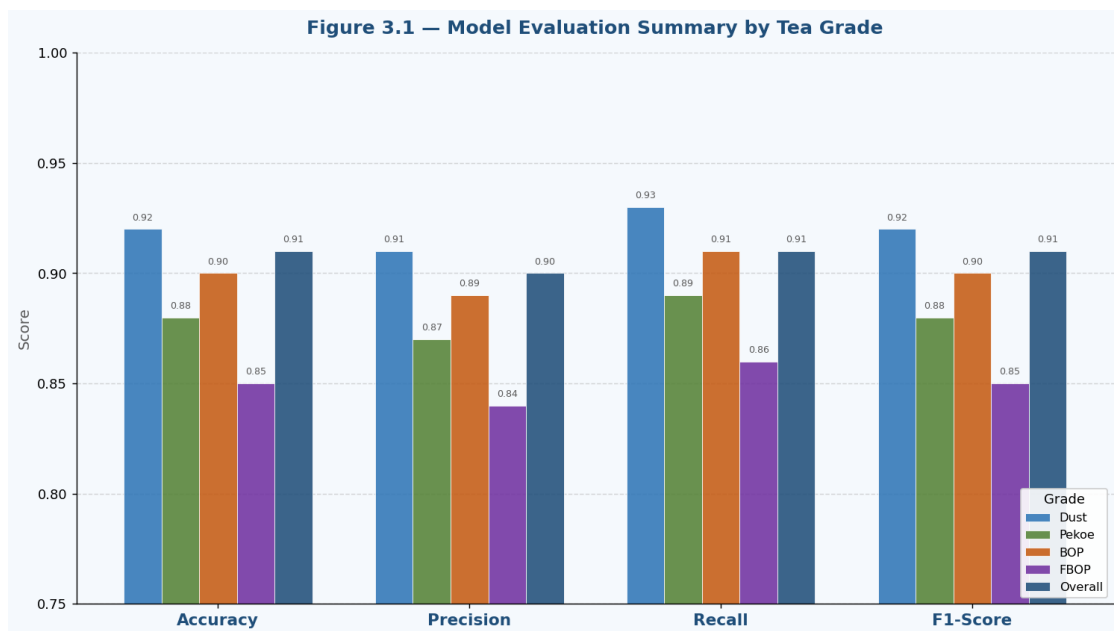


Figure 3.1 CNN Tea Grade Model Evaluation Summary

Figure 3.2 presents the confusion matrix for the AI tea grade classification model, illustrating the distribution of correct and incorrect predictions across all grade categories.

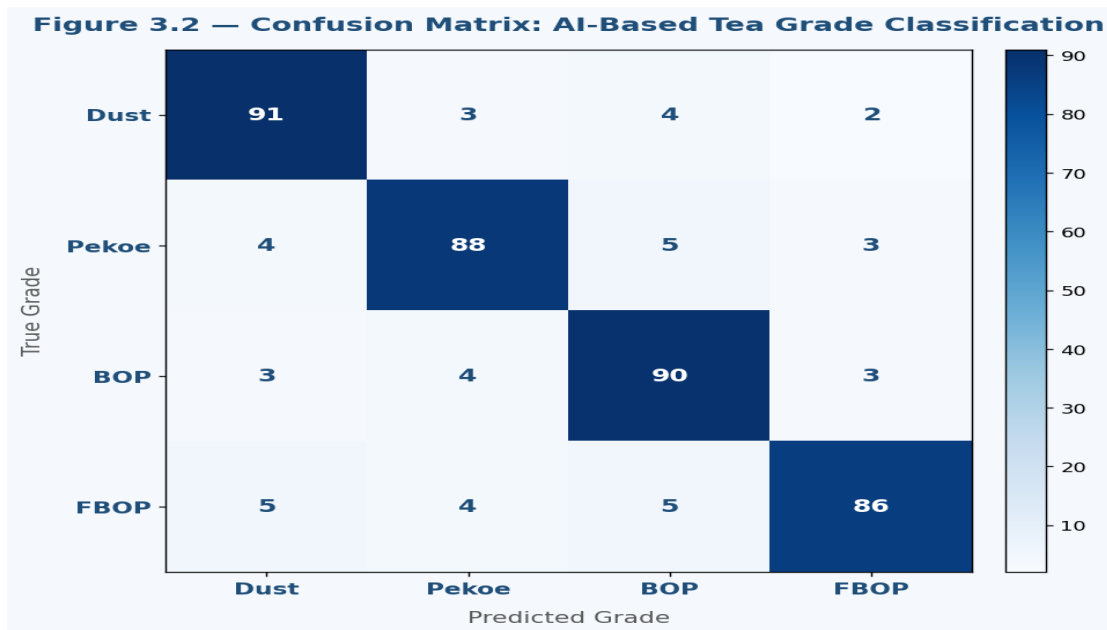


Figure 3.2 Confusion Matrix — AI Tea Grade Classification

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed System

The proposed system consists of six interconnected layers: image input, preprocessing, classification, comparison and validation, reporting, and decision support. Together these layers transform raw tea sample images into grade predictions and management-ready outputs.

The architecture is intentionally modular so that the dataset, CNN model, report logic, and user interface can be improved independently during future iterations.

4.2 System Components

Image Acquisition Module: captures or uploads tea sample images collected under standardized conditions.

Preprocessing Module: resizes images, normalizes color and lighting variation where possible, and prepares them for model input.

CNN Classification Engine: predicts the most likely grade label and associated confidence score for each image.

Manual Comparison Module: records manual grade labels and compares them against AI outputs for validation purposes.

Batch Analytics Module: aggregates predictions across a production batch and identifies dominant grades, distributions, and low-confidence cases.

Report Generation Module: produces batch-wise summaries, printable reports, and dashboard views for factory staff and management.

4.3 Component Interaction

The user uploads one or more tea sample images through the interface. The preprocessing module standardizes the images and forwards them to the CNN model. The classifier predicts the grade and confidence score. When manual labels are available, the comparison module calculates agreement and highlights mismatches. The analytics layer then prepares batch-level summaries which are presented through the reporting module.

4.4 Data Flow Explanation

Tea Image Input → Image Preprocessing → CNN Classification → Predicted Grade + Confidence Score → Manual Comparison (optional) → Batch Aggregation → Daily Report Generation → Decision Support Dashboard

4.5 Scalability and Security Considerations

The proposed architecture should allow additional grade categories and larger image datasets to be added in the future. Model retraining and version tracking should be supported so that performance can improve as more local data becomes available. Basic security measures should include authenticated access to the reporting interface, restricted access to administrative functions, and secure storage of image files and generated reports. If the system is extended for real factory deployment, backup policies and audit logs should also be included.

Figure 4.2 illustrates the high-level system architecture, showing the six interconnected layers that together transform raw tea sample images into grade predictions and management-ready outputs.

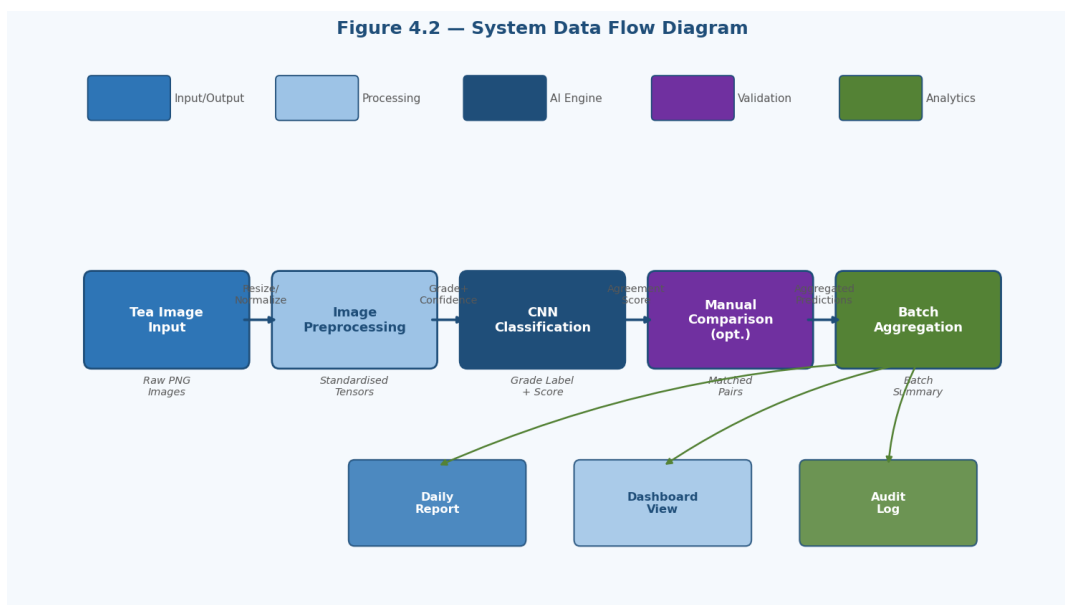


Figure 4.2 High-Level System Architecture

Figure 4.1 shows the system data flow diagram, tracing the movement of information from initial tea image input through preprocessing, classification, batch aggregation, and final report generation.

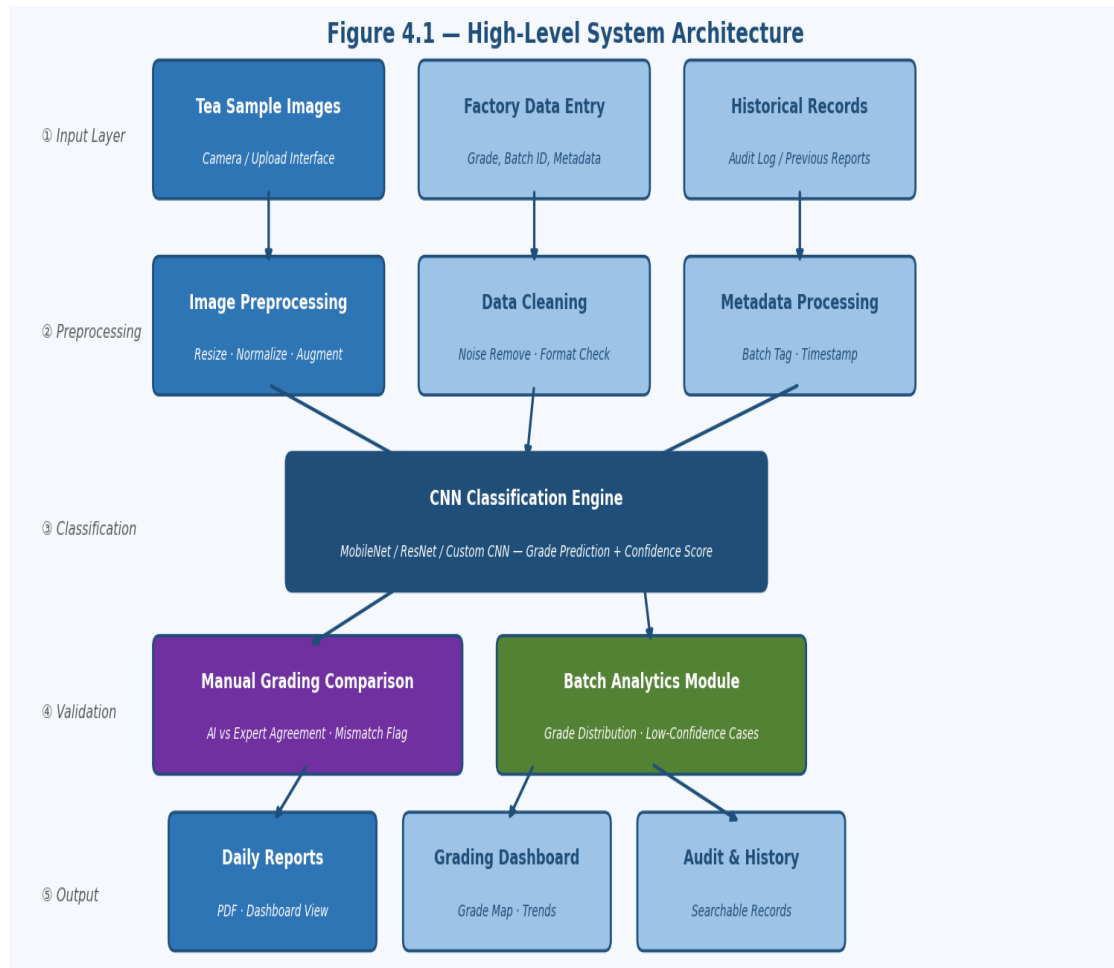


Figure 4.1 System Data Flow Diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed Tea Sorting and Grading Automation system involves several stakeholders who either use the system directly or benefit from its outputs.

Factory Managers are responsible for reviewing production quality, monitoring batch consistency, and making operational decisions. They use the system to review daily reports, grade distributions, and exception cases.

Quality Control Officers and Tea Graders are the main operational users. They upload images, review predicted grades, compare AI outputs with manual judgments, and use the batch summaries during quality checks.

Production Supervisors use the generated reports to monitor consistency across shifts and to identify batches that require closer inspection.

Researchers and System Developers use the system to manage datasets, retrain models, test alternative architectures, and evaluate grading accuracy under Sri Lankan conditions.

5.2 Functional Requirements

FR1: The system shall allow users to upload one or more tea sample images through a web-based interface.

FR2: The system shall preprocess uploaded images to improve consistency before model inference.

FR3: The system shall classify tea images into pre-defined tea grades using a trained CNN model.

FR4: The system shall display a confidence score together with each predicted grade.

FR5: The system shall allow manual grade labels to be entered for comparison and validation.

FR6: The system shall store image records, predicted grades, confidence scores, and related metadata in a database.

FR7: The system shall generate batch-wise grade summaries and daily production reports.

FR8: The system shall highlight uncertain or mismatched classifications that may require human review.

FR9: The system shall maintain historical classification records for future analytics and model improvement.

5.3 Non-Functional Requirements

Security: The system must ensure secure storage of uploaded images, report records, and user login information. Access to administrative dashboards should be restricted through authentication mechanisms.

Performance: The grading model should provide prediction results within a practical response time so that the tool remains useful during routine quality checks.

Usability: The interface must be simple and understandable for factory users who may not have an advanced technical background.

Reliability: The system should maintain stable behavior when processing multiple images and repeated batches.

Scalability: The architecture should support future expansion for more images, more grades, and more reporting features.

Maintainability: Model versions, dataset changes, and report templates should be manageable without redesigning the full system.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed system includes tea factories, regional plantation companies, tea processing centers, and tea quality assurance units in Sri Lanka. These organizations require more consistent grading, better reporting, and stronger digital records for quality monitoring.

The main customer persona is the factory manager or quality-control manager responsible for grading consistency, production reporting, and audit readiness. Such users require a reliable and easy-to-use system that can support operational decisions without disrupting existing factory workflows.

Secondary users include tea brokers, industrial researchers, training institutions, and sector-level support agencies that may benefit from standardized grading analytics and demonstrable quality evidence.

6.2 Value Proposition

The proposed system provides a practical AI-assisted grading solution for tea factories that need more consistent and traceable grading outcomes. Unlike purely manual processes, the system offers repeatable image-based evaluation, confidence-aware decision support, and structured daily reporting.

Key value propositions include faster and more consistent grading assistance, reduced subjectivity across operators, batch-wise digital records, low-cost prototype extensibility, and a clear comparison layer between AI output and human judgment.

6.3 Business Model Overview

The project can be commercialized as a subscription-based software service or as an on-premise deployment package for factories that prefer local data control. Additional revenue could come from setup, customization, staff training, and periodic model retraining services.

A pilot-first approach is suitable for the Sri Lankan market. Small-scale pilot deployments in selected factories can demonstrate consistency gains and build trust before wider adoption.

6.4 Go-to-Market Approach

Initial market entry can begin through pilot collaborations with tea factories or plantation companies willing to test the prototype under supervised conditions. Demonstration of batch reporting, audit trails, and agreement with expert graders can be used as the main adoption driver.

Partnership opportunities may exist with tea research groups, digital-agriculture programs, and sector stakeholders that promote modernization of tea manufacturing.

6.5 Pricing Strategy

A tiered pricing strategy can be implemented based on factory size, number of users, and reporting complexity.

Basic Plan – for small factories needing limited image uploads and core grade classification.

Standard Plan – for medium factories needing batch analytics, historical reports, and manual comparison features.

Enterprise Plan – for large factories requiring unlimited usage, custom dashboards, integration support, and periodic retraining.

6.6 Competitive Advantage

The proposed system offers several important advantages: it is tea-specific, designed for Sri Lankan factory conditions, combines classification with reporting, and supports human decision-making rather than demanding full workflow replacement.

Compared with generic inspection tools, the solution focuses directly on tea grading categories and local operational practicality. Compared with fully manual grading, it provides repeatability, confidence visibility, and digital evidence.

6.7 Intellectual Property Considerations

Potential intellectual property components include the labeled tea-grade dataset, trained CNN models, batch reporting logic, and the integrated software interface. If factory-specific data is used, ownership and permitted usage should be clearly defined through agreements.

If the project moves beyond research, data licensing, model retraining rights, and deployment restrictions should be documented carefully.

6.8 Cost Estimation

The estimated cost for developing and deploying the prototype is structured around four categories. Development infrastructure, including cloud GPU resources, web hosting, and database services, is estimated at Rs. 25,500. Data collection and preparation, covering factory visits, annotation effort, and image management, is estimated at Rs. 7,500. Tooling and operational expenses such as AI development tools, internet, and utilities are estimated at Rs. 6,500. A contingency reserve of Rs. 2,500 is included to cover unforeseen expenses. The combined total is approximately Rs. 53,500, as detailed in the budget table in Section 7.

For a commercial deployment beyond the prototype stage, costs would scale with the number of factories, the volume of image data processed, and support requirements. A subscription pricing model, as described in Section 6.5, is designed to cover ongoing hosting, maintenance, and periodic model retraining costs while remaining accessible to small and medium tea factories.

7. BUDGET AND JUSTIFICATION

Table 7.1 presents the estimated budget for the proposed system, detailing each cost category, the allocated amount in Sri Lankan Rupees, and the justification for each expenditure.

Table 7.1 Estimated budget table

Description	Total Amount (Rs.)	Justification
Cloud GPU / model training resources (6 months)	12,000	Required for CNN experimentation, tuning, and retraining.
Image dataset storage and backup services	4,500	To maintain raw images, labeled sets, and backup copies.
Web application hosting / VPS	6,000	For prototype deployment and testing with a browser interface.
Database hosting or managed DB instance	3,000	Stores classifications, reports, user actions, and metadata.
AI development tools and runtime support	3,500	Supports preprocessing, experimentation, and evaluation workflows.
Internet and utilities	3,000	Shared development connectivity and power cost allocation.
Transport for factory-oriented data collection	5,000	Travel cost for collecting tea sample images and validating labels.
Annotation and image preparation support	2,500	Labeling effort, naming conventions, and file organization.
Miscellaneous / contingency	2,500	For unexpected accessories, storage, or replacement needs.
Hardware (camera / lighting for image capture)	6,500	Controlled-condition image capture setup; USB camera or smartphone mount with consistent lighting rig for reproducible dataset collection.
Human effort (research, development, and validation)	5,000	Estimated allowance for student researcher time allocated to dataset preparation, model training, expert consultation, and evaluation activities.
Total Budget	53,500	

8. WORK BREAKDOWN STRUCTURE (WBS)

The work breakdown structure organizes all research tasks, timelines, and expected deliverables across the full project period. Table 8.1 presents the complete task schedule from literature review through to final documentation.

Table 8.1 Work Breakdown Structure

Task ID	Task Description	Timeline	Deliverable
T1	Conduct literature review on tea grading, machine vision, and CNN-based classification	March	Literature review report
T2	Collect tea-grade image dataset from factory-related environments	March – April	Tea-grade image dataset
T3	Label and preprocess images (resize, normalization, augmentation, quality check)	April	Preprocessed dataset
T4	Design baseline CNN architecture and experimental workflow	May	CNN design specification
T5	Train and tune grading model using labeled dataset	May – June	Trained grading model
T6	Evaluate model using accuracy, precision, recall, F1-score, and confusion matrix	June	Model evaluation results
T7	Develop manual comparison and batch analytics module	July	Validation and analytics module
T8	Create report generation and dashboard interface	August	Reporting prototype
T9	Integrate classifier with web application workflow	September	Integrated prototype
T10	Perform testing with new samples and verify practical usability	October	Testing and validation report
T11	Prepare final documentation and presentation	November	Final research report

The Work Breakdown Structure (WBS) illustrates the hierarchical organization of the **AI-Based Tea Grading and Quality Classification System** project. It divides the project into major components such as research, model development, analytics modules, system integration, and testing. Each component is further broken down into sub-modules and tasks (T1–T11) to clearly define the workflow.

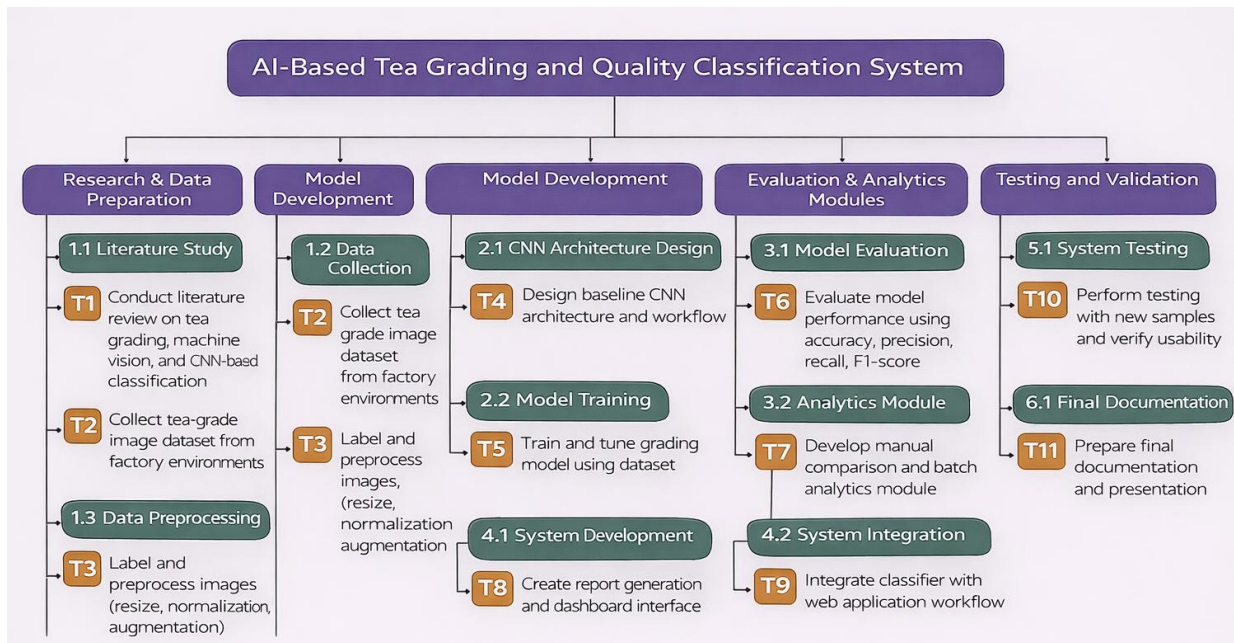


Figure 8.1 Work Breakdown Structure

9. GANTT CHART

The project timeline is illustrated in Figure 9.1, which presents the Gantt chart showing the schedule for all eleven research tasks across the project period from March to November 2026.

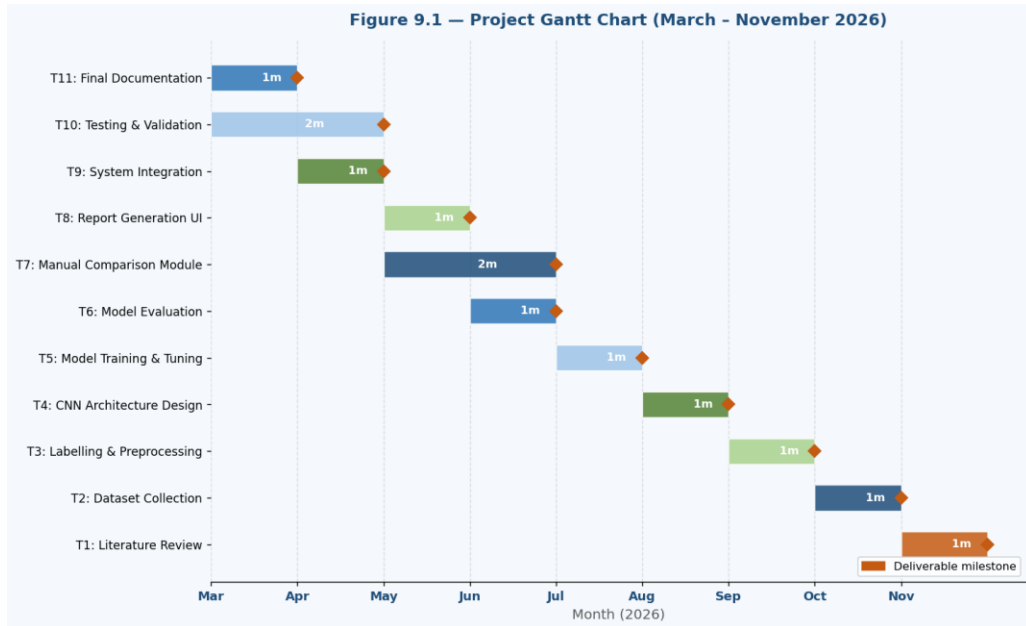


Figure 9.1 Project Gantt Chart (March–November 2026)

As shown in Figure 9.1, the project follows a sequential and partially overlapping schedule, with data collection and preprocessing tasks in the early phase, model development and evaluation in the middle phase, and system integration, testing, and documentation in the final phase.

10. REFERENCES LIST

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11. APPENDICES

11.1 Survey Instrument

The following instrument can be used to gather feedback from factory managers, graders, and quality control officers during the requirement and evaluation stages.

Section A – Respondent Information

1. What is your role in the tea factory or quality-control process?
2. How many years of experience do you have in tea grading or factory supervision?
3. How often do you participate in grade verification or production reporting?

Section B – Current Grading Practice

4. How are tea grades currently confirmed in your workplace?
5. What visual characteristics are most important when identifying a grade?
6. What are the most common difficulties when grading borderline batches?

Section C – Perceived Challenges

7. How often do manual grading outcomes vary between operators?
8. Which causes are most responsible for inconsistent grading?
9. How useful would a confidence-aware AI suggestion be during grading?

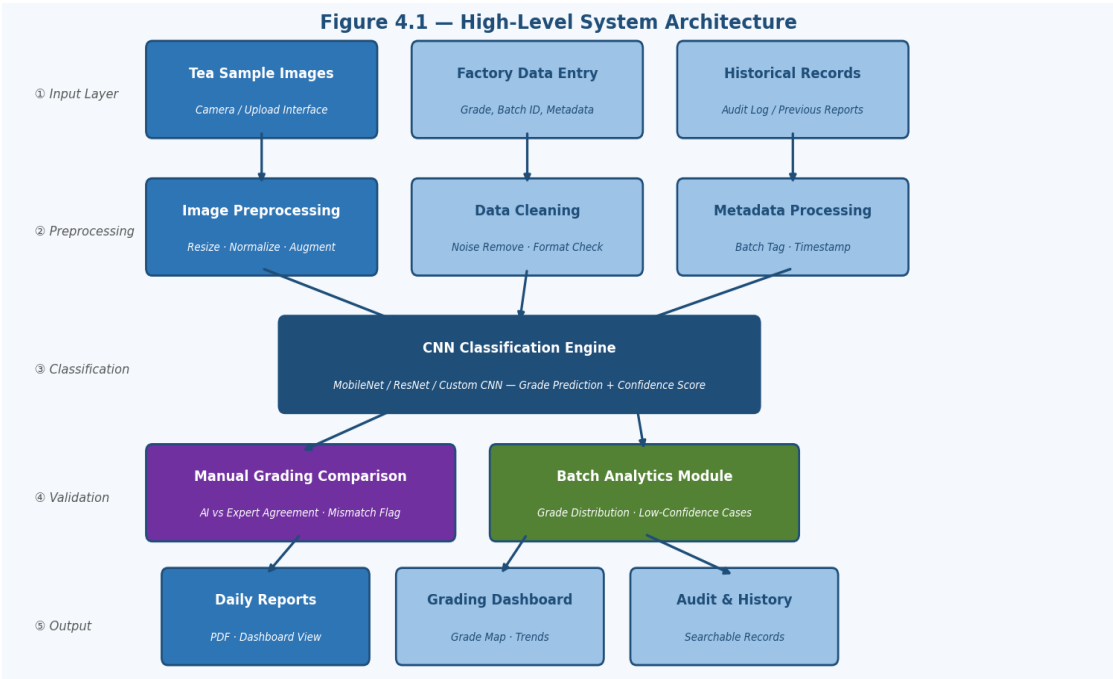
Section D – System Evaluation

10. Was the predicted grade understandable and relevant?
11. Did the batch report provide useful information for review?
12. Would you use this system as a decision-support tool in routine operations?

11.2 Detailed diagrams

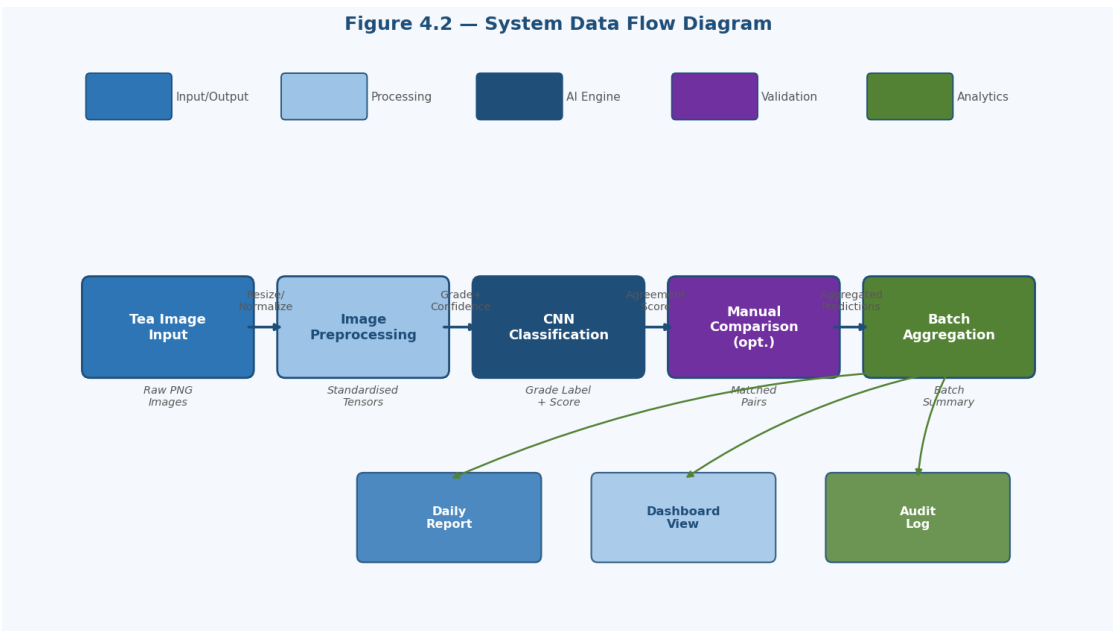
The following diagrams present the detailed technical architecture and data flow of the proposed AI-Based Tea Sorting and Grading Automation system.

Appendix Figure A1 presents the detailed system architecture diagram, showing the complete layered structure of the proposed AI-based tea grading system including all major modules and their interconnections.



Appendix Figure A1. Detailed System Architecture Diagram

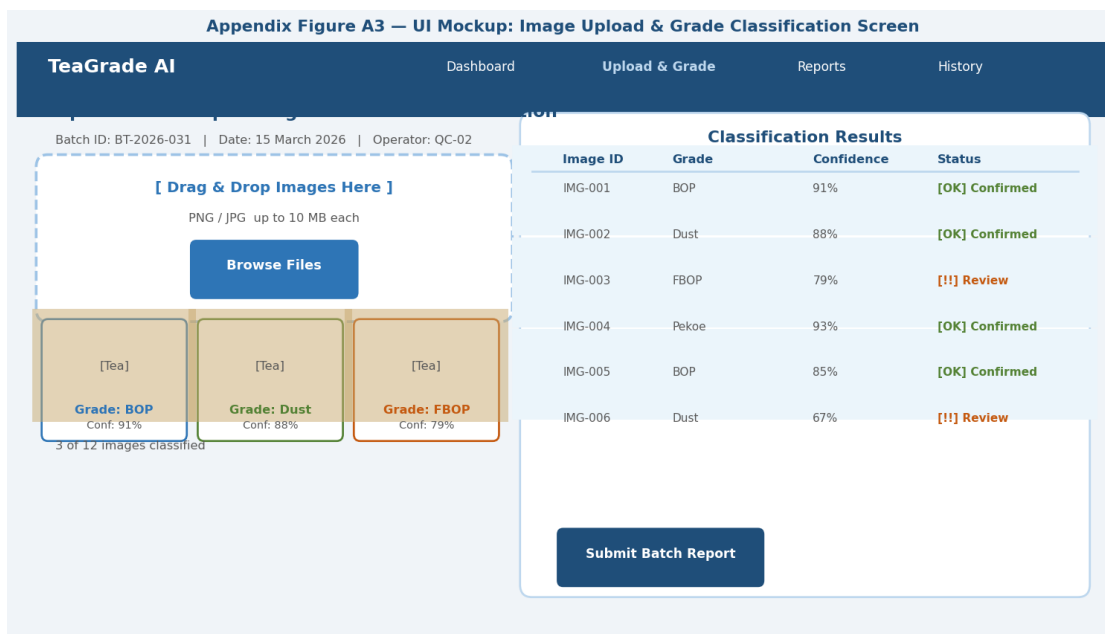
Appendix Figure A2 presents the detailed data flow diagram, tracing step-by-step how data moves through the system from image input to final decision support output.



Appendix Figure A2. Detailed Data Flow Diagram

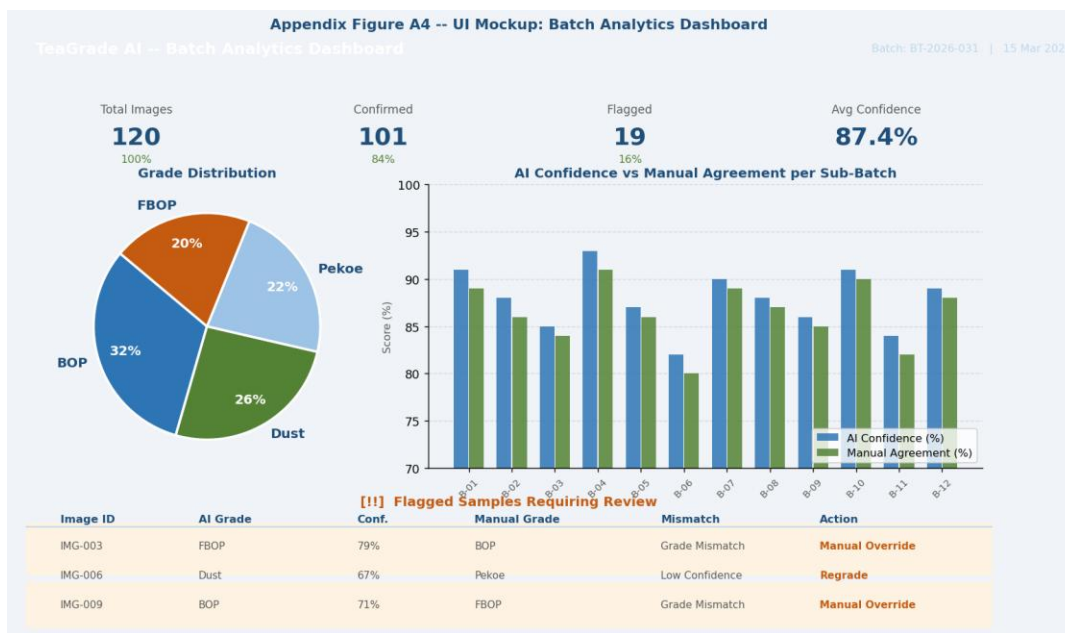
11.3 Preliminary UI mockups

The following UI mockups illustrate the key screens of the proposed prototype interface for the AI-based grading system. Appendix Figure A3 shows the UI mockup for the image upload and grade classification screen, depicting how users submit tea sample images and receive predicted grade results with confidence scores.



Appendix Figure A3. UI Mockup — Image Upload and Grade Classification Screen

Appendix Figure A4 shows the UI mockup for the batch analytics dashboard, illustrating how grade distribution summaries, production statistics, and exception flags are displayed for factory managers.



Appendix Figure A4. UI Mockup — Batch Analytics Dashboard

11.4 Risk analysis

This section identifies the key risks associated with the proposed research and provides mitigation strategies for each. Table 11.1 summarizes the identified risks, their potential impact, likelihood of occurrence, and corresponding mitigation approaches.

Table 11.1 Risk Analysis Table

Risk	Impact	Likelihood	Mitigation Strategy
Insufficient image diversity across grades	High	Medium	Collect samples from multiple batches and conditions; expand dataset gradually.
Incorrect labels from manual verification	High	Medium	Use expert review and double-check labeling for uncertain samples.
Model overfitting on small dataset	High	Medium	Apply augmentation, validation split, and compare transfer-learning baselines.
Lighting variation affecting image quality	Medium	High	Use controlled background and standardized capture guidelines.
Low user trust in AI results	High	Medium	Show confidence scores and manual-versus-AI comparison in reports.
Delayed access to factory data collection	Medium	Medium	Use phased data acquisition and reserve secondary sample collection plan.

As presented in Table 11.1, the six identified risks span dataset quality, model overfitting, image capture conditions, user trust, and factory access challenges. Each risk is rated for impact and likelihood, and an appropriate mitigation strategy is assigned to ensure the research can be completed successfully within the project scope.

11.5 Ethical documents

Appendix Note 1: Permission letter template for factory-oriented image collection.

Appendix Note 2: Participant information sheet for expert feedback or grading validation.

Appendix Note 3: Data handling note describing that uploaded sample images and grading records will be used only for approved academic research purposes.

Appendix Note 4: Confidentiality statement for any factory-specific information, sample naming, and production-related observations.